

Arunabha Majumder

Test Engineer | Verification & Validation | Test Protocols | Data Analysis | Assistive Devices

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Portfolio: arunabha22.github.io/arunabha-majumder | Project: www.viexo.aau.dk

PROFILE

Test and verification engineer with a BTech in Mechanical Engineering, an MTech in Mechatronics and a PhD in progress at Aalborg University. Plans, coordinates and reports testing and verification activities in product development projects: writing test protocols, designing test set-ups, running controlled tests, analysing data and communicating conclusions to stakeholders. Experienced with assistive and rehabilitation devices tested with human participants, including statistical study design and ethics protocols. Analytical, detail-oriented and collaborative across disciplines; fluent in English (C1).

KEY SKILLS

- **Verification & Validation:** Test planning, test protocols, controlled test conditions, verification against requirements and reference measurements, validation of measurement methods
- **Data Analysis:** Statistical analysis, statistical power justification, data pipelines, interpreting results and drawing conclusions (Python, MATLAB)
- **Test Equipment & Instrumentation:** Custom test rigs and fixtures, force/torque sensors, encoders, DAQ, EMG, motion capture (Qualisys)
- **Device Testing:** Human-subject test studies, ethics protocols, assistive and rehabilitation devices
- **Communication:** Reporting to technical and non-technical stakeholders, cross-functional collaboration, teaching and supervision, peer-reviewed publications
- **Engineering Tools:** SolidWorks (CSWA), Autodesk Fusion 360, FEA, prototyping

PROFESSIONAL EXPERIENCE

Research Engineer – Device Development, Testing & Verification (PhD Project: VIE XO) | 2023 – Present

Aalborg University, Aalborg, Denmark

- Planned, coordinated and reported a 12-participant verification study of a shoulder exoskeleton across 3 operating modes and 2 load conditions, covering statistical power justification, test protocol, ethics protocol, EMG and motion-capture instrumentation, and the end-to-end data pipeline.
- Validated an end-effector force-estimation method on a 5-bar planar parallel robot, using variable-stiffness actuators and the robot Jacobian, against a reference force sensor to below 10% RMS error.
- Designed custom automated test rigs and ran controlled tests on prototype hardware with motor, encoder, force sensors and DAQ.
- Showed through testing that a hybrid actuation design cut energy consumption by more than 25% and an Assist-as-Needed controller reduced user muscle effort by more than 15% (Best Paper Award, IFToMM ISRM 2026).
- Communicated results in peer-reviewed publications; taught university students and supervised Bachelor's (B.Tech) students (2023 – 2025).

MTech Researcher – Pneumatic Actuator Testing & Characterisation (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India

- Built a test set-up with valves, pressure sensors and DAQ, and characterised the load–displacement behaviour of in-house pneumatic actuators to derive an empirical model.
- Verified a neural-network-based controller against a classical PID baseline under varying load; tracking error of 0.3–0.78%.

Mechanical Engineering Capstone – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019

Siddaganga Institute of Technology, Tumakuru, India

- Led design, fabrication and user testing of a rehabilitation device; won the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant.

EDUCATION

PhD Candidate, Mechatronics and Robotics | 2023 – Present

Aalborg University, Department of Materials and Production, Denmark

MTech, Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), India

BTech, Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), India

PUBLICATIONS, CERTIFICATIONS AND AWARDS

- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, MMS, vol. 213, Springer. Best Research Paper Award.
- Majumder, A., et al. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.
- Certified SolidWorks Associate (CSWA); TATA Technologies Innovation Award (2019)

TECHNICAL TOOLS

- **Data & Analysis:** Python, MATLAB/Simulink, LabVIEW (basic)
- **Test & Measurement:** Qualisys Motion Capture, Forsentec Force/Torque Sensors, EMG, DAQ, custom test rigs
- **CAD & Prototyping:** SolidWorks, Autodesk Fusion 360, FEA, 3D printing

LANGUAGES

English (C1, fluent) | Bengali (Native) | Hindi (Conversational)

REFERENCES

Available upon request.